

WiP: Sparse Dual-Rate Vision for Long-Term Coral Reef Monitoring

Sitong Shen

The University of Hong Kong
Hong Kong SAR, China
shenst@connect.hku.hk

Chenshu Wu

The University of Hong Kong
Hong Kong SAR, China
chenshu@cs.hku.hk

Abstract

Long-term coral reef monitoring is essential for understanding underwater ecosystem health. However, dominant monitoring approaches often rely on infrequent diver surveys and spatially coarse satellite sensing. While underwater cameras have recently emerged as fixed sensing nodes for persistent monitoring, conventional systems treat each observation as a full, independent frame. This causes resource usage to scale with frame volume rather than ecological change. To address this inefficiency bottleneck, we explore an edge-shore collaborative architecture for continuous, long-term visual monitoring under ultra-low edge power constraints. We observe that fixed-view coral monitoring contains multi-rate visual dynamics and substantial local temporal redundancy. Specifically, monochrome captures short-term variation, while color and appearance can be refreshed more sparsely, leaving many local regions unchanged over time. Inspired by these insights, we design a dual-camera edge-shore system that keeps sparse RGB anchoring, frequent monochrome sensing, temporal reuse, and compact updates on the edge, while offloading expensive ecological vision tasks to the shore. Early prototyping on STM32-class MCUs demonstrates that our approach significantly reduces underwater sensing, computation, and I/O overhead while fully preserving information for downstream coral-related tasks.

1 Introduction

Coral reefs are irreplaceable ecological infrastructure: they sustain marine biodiversity, support worldwide ecosystems, and provide natural coastal protection [1]. However, recurrent bleaching and large-scale mortality are accelerating reef degradation [2]. Long-term monitoring is therefore essential, as reef decline and recovery unfold over years to decades and cannot be captured by isolated observations [3].

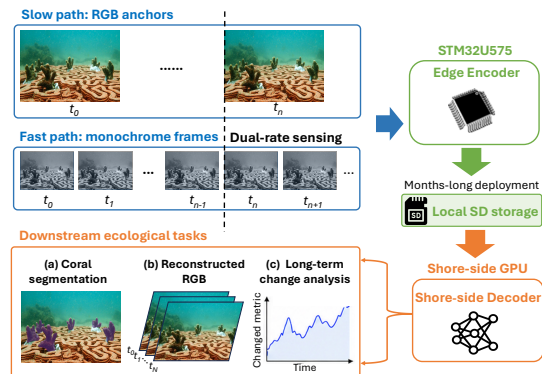


Figure 1: High-level architecture of the system.

Existing coral reef monitoring approaches remain difficult to scale in temporal continuity, spatial resolution, and deployment density: (1) diver surveys provide detailed in situ observations but are labor-intensive and infrequent, typically repeated only annually or on multi-year cycles [4]; (2) satellite remote sensing offers broad coverage but cannot resolve colony-level changes [5]; and (3) underwater monitoring platforms, including fixed camera stations and autonomous underwater vehicles, provide high-resolution observations but remain costly for dense, long-term deployment [6].

Low-power visual sensing platforms offer a scalable path to persistent underwater observation, but long-term deployments remain energy-limited. Because continuous underwater data transfer is costly and energy-intensive, existing deployments often store observations locally for later retrieval [6]. Yet conventional systems still process and write each frame independently, wasting resources when fixed-view reef scenes change slowly. Our key insight is to make underwater resource usage follow newly observed visual changes rather than full-frame processing.

We present an early prototype of an offline edge-shore coral monitoring system for resource-constrained underwater nodes. Figure 1 illustrates the high-level architecture. The system combines sparse dual-rate sensing, temporal feature reuse, and shore-side reconstruction. The node captures sparse RGB anchors and frequent monochrome observations, updates newly observed content, and stores compact representations locally. After retrieval, a shore-side model reconstructs visual observations and supports ecological analysis.



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MobiCom '26, Austin, TX, USA

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ACM ISBN 979-8-4007-2505-0/26/10

<https://doi.org/10.1145/3795866.3844768>

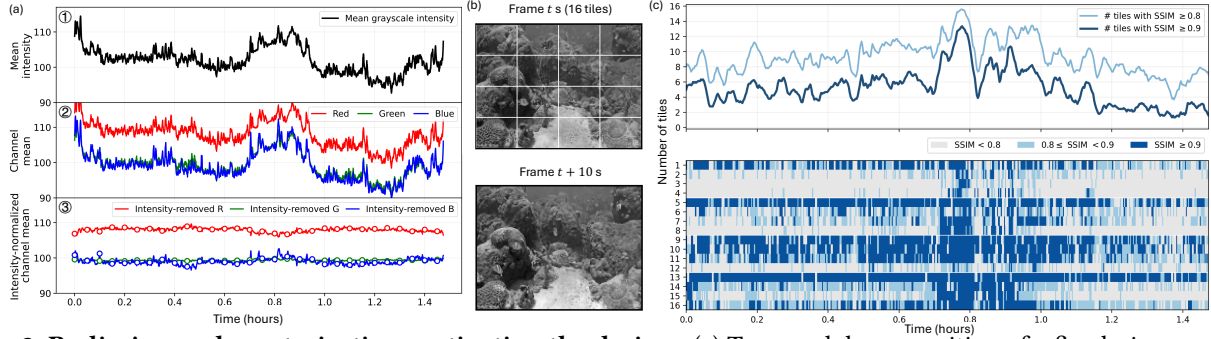


Figure 2: Preliminary characterization motivating the design. (a) Temporal decomposition of a fixed-view recording into a1) global grayscale intensity, a2) raw RGB channel means, and a3) RGB variation after removing global intensity. (b) Example observations sampled 10 s apart with a 4×4 tile grid. (c) Tile-level temporal similarity over time.

Preliminary results on an 88-min fixed-view Coral City Camera video [7] show strong tile-level redundancy, with 32.6% of 10-s tile pairs exceeding 0.90 structural similarity (SSIM). Power measurements show that microSD writing consumes ~ 330 mW, over $22\times$ camera/MCU power. These results indicate opportunities to reduce redundant edge computation and storage while preserving shore-side analysis utility.

2 Motivation and Design Rationale

Long-term coral monitoring requires dense visual records that capture gradual degradation, recovery, and structural changes at the colony scale. However, maintaining such records on underwater nodes for months to years is fundamentally constrained by limited energy availability. Unlike terrestrial sensing systems, underwater deployments cannot rely on frequent maintenance, making repeated sensing, computation, and storage operations critical factors that determine deployment lifetime.

Challenge 1: Energy-efficient acquisition and local storage: Continuous high-resolution RGB imaging provides rich visual information but incurs substantial sensing, memory-transfer, and local-storage overhead. Repeatedly writing large image streams to onboard storage can become a major contributor to autonomous sensing energy [8, 9]. The system must therefore preserve temporal coverage without continuously acquiring and writing expensive visual observations. This motivates sparse dual-rate sensing, where infrequent RGB frames provide visual anchors and lightweight monochrome observations track short-term scene evolution.

Challenge 2: Long-horizon representation update: Fixed-view coral scenes often evolve slowly, so consecutive observations contain substantial spatial and temporal redundancy. Yet conventional vision pipelines process and store each frame independently, making resource use scale with frame volume rather than newly observed information. The challenge is to retain meaningful visual changes while preserving information for shore-side reconstruction and ecological analysis. This motivates temporal reuse, which updates

changed regions and stores compact representations instead of complete observations.

Challenge 3: End-to-end system feasibility: Hardware-software co-design under tight power constraints and real underwater conditions remains a system challenge.

3 System Design

Figure 1 presents the architecture of the proposed offline edge-shore system. The system consists of an underwater node and a shore-side processing pipeline. The underwater node executes dual-rate sensing and selective representation updates, reducing edge computation and local storage costs; the shore side performs reconstruction and ecological analysis after data retrieval.

Sparse dual-rate sensing: The underwater node captures sparse high-resolution RGB frames as visual anchors and frequent low-resolution monochrome frames for temporal tracking. RGB anchors preserve color and appearance information required for later reconstruction [10], while monochrome observations provide dense temporal context at lower sensing and storage cost. This dual-rate design avoids continuously acquiring high-cost RGB images while retaining the visual cues needed for shore-side analysis.

Temporal representation reuse: Rather than encoding every frame independently, the node maintains visual states and updates only regions with new information. Unchanged regions reuse prior representations, while changed regions are stored as compact updates with metadata. After retrieval, the shore-side model decodes the stored data for RGB reconstruction, coral segmentation, and long-term change analysis. To bound long-horizon drift, the node maintains a reuse age map and tracks global residual accumulation, triggering dense refreshes when states become stale.

4 Preliminary Results

We analyze an 88-min fixed-view coral video from Coral City Camera, a public reef-camera platform [7], sampled every 10 s to match our cadence. It assesses three design choices: dual-rate sensing, adaptive reuse, and compact local storage.

Temporal intensity and color dynamics: We analyze a fixed-view coral recording and normalize each frame to the same global grayscale intensity, preserving only changes in the relative RGB composition. Fig. 2(a) shows that RGB variation after removing global intensity is much narrower than raw RGB variation, suggesting that frequent monochrome observations can track short-term dynamics while sparse RGB anchors preserve color and appearance for shore-side reconstruction.

Tile-level redundancy: We compare consecutive fixed-view observations sampled 10 s apart using tile-level structural similarity (SSIM). Each frame is divided into a 4×4 tile grid, yielding 8,496 corresponding tile pairs across the analyzed sequence. The median tile SSIM is 0.834, with 32.6% of tile pairs above 0.90 and 69.9% of intervals containing at least four tiles above 0.90. Fig. 2(b)–(c) further shows that these high-SSIM regions persist but vary substantially across both space and time. This motivates adaptive tile-level updates rather than full-frame recomputation or a fixed reuse policy.

Storage cost: Table 1 shows that microSD write power is much higher than the active power of the camera sensing and compute components considered here. Although total energy also depends on duty cycle and write duration, this motivates compact updates that avoid repeatedly writing redundant complete observations.

5 Discussion and Future Work

This work positions long-term coral monitoring as a resource-aware underwater visual sensing problem: the node should preserve visual evidence for scientific analysis while avoiding repeated acquisition, computation, and storage of unchanged content. Our preliminary characterization suggests that fixed-view reef scenes provide this opportunity, as visual signals evolve at different rates and local regions exhibit persistent temporal similarity. These observations support a broader design direction in which underwater visual monitoring systems adapt their resource use to scene evolution rather than raw capture volume.

Future work will move from prototype validation toward field-deployable reef monitoring systems. We will study how dual-rate sensing, temporal reuse, and edge–shore processing behave over longer deployments, across changing illumination, water clarity, biological motion, and seasonal reef dynamics. We will also extend the shore-side pipeline beyond reconstruction and segmentation toward richer ecological interpretation, including long-term change detection, bleaching assessment, and restoration monitoring.

6 Conclusion

This work presents an early offline edge–shore vision prototype for long-term coral reef monitoring on resource-constrained underwater nodes. By combining sparse dual-rate sensing,

Table 1: Power measurement of key components.

Component	Operating mode	Active current	Active power
HM01B0	Fast mono, QQVGA(30 FPS)	–	0.67 mW
HM0360	Slow RGB, VGA(60 FPS)	7 mA	~ 12.6 mW
STM32U575	Active compute(160 MHz)	8.25 mA	14.85 mW
SanDisk SD	Standard-mode write	~ 100 mA	330 mW

temporal representation reuse, and shore-side reconstruction, the proposed system reduces redundant underwater processing and storage while preserving information for downstream ecological analysis. Preliminary fixed-view video characterization, active-power measurements, and STM32-class prototyping indicate the feasibility of this direction toward persistent, low-maintenance underwater coral monitoring systems for long-horizon scientific observation.

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